

What is Penetration Testing and Why is it Important?

Penetration testing, also known as pen testing or security testing, is a form of ethical hacking. It's an authorised cyber attack on any computer system, which is carried out to evaluate the security of that system. In this article we will see why penetration testing is so important for organisations.



Penetration testing is done with the sole purpose of identifying the vulnerabilities within a system and fixing them. This ensures that unauthorised third parties don't access the data within the system. Such testing helps to get a full risk assessment of any system, including its strengths and weaknesses. The process usually begins with identifying the target system as well as a particular goal after this system is tested. The information received after testing the system is evaluated against the project's initial plan. Security issues that are uncovered in this manner are reported to the system owner.

With the rise of cyber attacks, organisations must perform regular pen tests to secure themselves from data breaches and to protect their users' sensitive information.

What is a penetration test?

A penetration test, also called a pen test, is a planned cyber attack on your computer system to check for vulnerabilities. This is done to evaluate the security of the system. The UK National Cyber Security Center (<https://www.ncsc.gov.uk/pdfs/guidance/penetration-testing.pdf>) describes penetration testing as: "A method for gaining assurance in the security of an IT system by attempting to breach some or all of that system's security, using the same tools and techniques as an adversary might."

Pen testing is a form of ethical hacking done by white hat testers. They ensure that the computer and the data remain secure from hackers. The concept of penetration testing was born in the mid-1960s, as time-sharing

computer systems grew popular. These systems made resources available over communication lines, but also created new security concerns. With 640 terabytes of data tripping around the globe every minute these days, there's a massive amount of data that hackers can steal. The key to not being on the receiving end of one of these attacks is ensuring that your security system is up to date. Conducting a thorough penetration test will ensure that potential vulnerabilities are plugged and a data breach is avoided.

How are such tests done?

Penetration testing can be used to test a network, Web application, or a computer system to find potential vulnerabilities that can be exploited by a hacker. It can be a process, tool,